Amigurumi Pattern Logic

Technical Research for Automated Pattern Generation

1. Construction Fundamentals

Amigurumi creation relies on **volumetric stitch manipulation**. Unlike flat crochet, amigurumi uses a "spiral" round technique to eliminate visible seams and create a continuous fabric texture.

- **Single Crochet (sc):** The primary building block. Stitches must be tight to prevent stuffing (poly-fill) from leaking.
- **Staggered Increases:** To maintain perfect circles, increases must not be stacked directly above each other, or the piece will become polygonal (a hexagon).

2. Geometric Mathematics

Flat Circle Logic

The standard increase rate for a flat disc starting with 6 stitches:

$$Stitches_n = 6 \times n$$

3D Surface Curvature

For a sphere, the stitch count per round follows a sine-based growth curve:

$$C \propto \sin(\theta)$$

Where θ is the polar angle. An app algorithm must calculate the derivative of the shape's profile to determine the local increase/decrease rate.

3. Application Logic Workflow

A. Shape Decomposition

Identify the silhouette and break the body down into **Primitive Objects**: Head (Sphere), Torso (Ovoid), Limbs (Cylinders).

B. Stitch Mapping

Convert the 3D surface area into a 2D instruction set. Algorithms must account for the **Stitch Aspect Ratio**: Single crochet stitches are roughly 20% wider than they are tall.

4. Advanced High-Fidelity Factors

- **Color Transitions:** "Fair Isle" vs. "Tapestry" methods for mapping image pixels to yarn colors.
- **Structural Tension:** Incorporating "short rows" to create bends (elbows/knees) without separate pieces.